❖ **Step-3** Square each deviations of 'X' and 'Y' then obtain the sum of the Squared deviation, That is and

❖ **Step-4** Multiply each deviation under X with each deviation under Y and obtain the product of 'xy'. Then obtain the sum of the product of X,Y. Then obtain the sum of the product of x,y is Σxy.

❖ **Step-5** Substitute the value in the formula.

Formula of Karl Pearson's Coefficient of Correlation - Ungrouped Data.

Direct Method	Indirect Method
Using the Variables as it is	Actual Mean deviation method

1 $$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}}$$

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2}\sqrt{N\Sigma Y^2 - (\Sigma Y)^2}}$$

where $x = (x-\bar{x})$, $y = (y-\bar{y})$

2. Assumed Mean Deviation Method

$$r = \frac{N\Sigma dxdy - (\Sigma dx)(\Sigma dy)}{\sqrt{N\Sigma dx^2 - (\Sigma dx)^2}\sqrt{N\Sigma dy^2 - (\Sigma dy)^2}}$$

2. Indirect Method

$dx = (x-\bar{x})$ and $dy = (y-\bar{y})$

r is free from origin

r is free from unit of measurement $-1 \le r \le +1$

Direct Method:

Example 1: Calculate Karl Pearson's Coefficient of correlation from the following data and interpret its value:

Price :X	10	12	14	15	19
Supply:Y	40	41	48	60	50

Solution: Let us take Price as X and supply as Y

Computation of Pearson's Correlation Coefficient				
Price: X	Supply: Y	XY	X^2	Y^2
10	40	400	100	1600
12	41	492	144	1681
14	48	672	196	2304
15	60	900	225	3600
19	50	950	361	2500
$\Sigma x = 70$	$\Sigma y = 239$	$\Sigma xy = 3414$	$\Sigma x^2 = 1026$	$\Sigma y^2 = 11685$

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2}\sqrt{N\Sigma Y^2 - (\Sigma Y)^2}} \qquad r = \frac{(5 \times 3414) - (70 \times 239)}{\sqrt{(5 \times 1026) - (70)^2}\sqrt{5x11685 - (239)^2}}$$

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